

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 6

[illegible]

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^{\dagger}D_{\mu_1}\left(S_{1,j_1}\right)$	$+\frac{2}{3}h\log\left(\frac{\mu\hbar\alpha^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}S_{1,j_2}^2y_{1,i_1,\tau_2}y_{1,i_2,\tau_1}}}\left(S_{1,j_1}\right)^2$ $-\frac{2C\text{NewSera}r12h\log\left(\frac{\mu\hbar\alpha^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,\tau_1}S_{1,j_1}y_{1,i_1,\tau_1}S_{1,j_1}^{\dagger}}}$ $-\frac{\frac{4}{3}h\log\left(\frac{\mu\hbar\alpha^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,\tau_2}y_{1,i_2,\tau_1}S_{1,j_1}^{\dagger}S_{1,j_2}^{\dagger}}}$ $-\frac{2h\log\left(\frac{\mu\hbar\alpha^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,\tau_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,\tau_1}S_{1,j_1}^{\dagger}S_{1,j_2}^{\dagger}I_{\mathbb{Z}1,j_3,j_1,j_4,j_2}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}$ $-\frac{2h\log\left(\frac{\mu\hbar\alpha^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,\tau_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,\tau_1}S_{1,j_1}^{\dagger}S_{1,j_2}^{\dagger}I_{\mathbb{Z}1,j_3,j_1,j_4,j_2}^{-2}\lambda_{\mathbb{Z}1}^{-2}}}$	$+\frac{2\hbar\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}I_{\mathbb{Z}2,j_3}^{-1}}}{\sqrt{6}M_F}e^{-T}C\overline{P_2}I_{\mathbb{Z}2,j_1}H_{\mathbb{Z}1}H_{\mathbb{Z}2}\lambda_{\mathbb{Z}1}\text{Bar}(\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L}))_{\lambda_{\mathbb{Z}1},\omega_2}\text{Bar}(\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L}))_{\lambda_{\mathbb{Z}2},\omega_3}$ $+\frac{2\hbar\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}I_{\mathbb{Z}2,j_3}^{-1}}}{\sqrt{6}M_F}e^{-T}C\overline{P_2}I_{\mathbb{Z}2,j_1}H_{\mathbb{Z}1}H_{\mathbb{Z}2}\lambda_{\mathbb{Z}1}\text{Bar}(\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L}))_{\lambda_{\mathbb{Z}1},\omega_2}\text{Bar}(\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L}))_{\lambda_{\mathbb{Z}2},\omega_4}$ $+\frac{2\hbar\overline{y_{1,i_1,\tau_1}C\overline{P_2}I_{\mathbb{Z}2,j_1}I_{\mathbb{Z}2,j_2}^{-1}}}{\sqrt{3}M_F}\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L})_{\lambda_{\mathbb{Z}1},\omega_3}\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L})_{\lambda_{\mathbb{Z}2},\omega_4}}}\overline{H_{\mathbb{Z}1}^{-1}H_{\mathbb{Z}2}^{-1}}$ $+\frac{2\hbar\overline{y_{1,i_1,\tau_1}C\overline{P_2}I_{\mathbb{Z}2,j_1}I_{\mathbb{Z}2,j_2}^{-1}}}{\sqrt{3}M_F}\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L})_{\lambda_{\mathbb{Z}1},\omega_3}\text{epn}(\mathbb{S}\mathbb{U}2\mathbb{L})_{\lambda_{\mathbb{Z}2},\omega_4}}}\overline{H_{\mathbb{Z}1}^{-1}H_{\mathbb{Z}2}^{-1}}$ $-\frac{2\hbar\overline{y_{1,i_1,\tau_1}I_{\mathbb{Z}2,j_3}^{-1}}}{\sqrt{6}M_F}\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}Y_{\mathbb{Z}1,j_3}H_{\mathbb{Z}1}^{-1}S_{1,j_3}^{-1}}}$ $-\frac{2\hbar\overline{y_{1,i_1,\tau_1}I_{\mathbb{Z}2,j_3}^{-1}}}{\sqrt{6}M_F}\frac{1}{\overline{y_{1,i_1,\tau_1}y_{1,i_2,\tau_2}Y_{\mathbb{Z}1,j_3}H_{\mathbb{Z}1}^{-1}S_{1,j_3}^{-1}}}$ $-\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,\tau_1}Y_{\mathbb{Z}2,j_3}^{-1}}}{\epsilon}\frac{1}{\overline{H_{\mathbb{Z}2}S_{1,j_2}y_{1,i_2,\tau_2}Y_{\mathbb{Z}1,j_1}H_{\mathbb{Z}1}^{-1}S_{1,j_1}^{-1}I_{\mathbb{Z}1,j_1,j_2,j_3,j_4}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}$ $-2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,\tau_1}Y_{\mathbb{Z}2,j_3}^{-1}}\frac{1}{\overline{H_{\mathbb{Z}2}S_{1,j_2}y_{1,i_2,\tau_2}Y_{\mathbb{Z}1,j_1}H_{\mathbb{Z}1}^{-1}S_{1,j_1}^{-1}I_{\mathbb{Z}1,j_1,j_2,j_3,j_4}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}$ $+\frac{1}{\overline{H_{\mathbb{Z}2}S_{1,j_2}H_{\mathbb{Z}1}^{-1}S_{1,j_1}^{-1}I_{\mathbb{Z}1,j_1,j_2,j_3,j_4}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}$ $-\frac{2\hbar\overline{y_{1,i_1,\tau_1}H_{\mathbb{Z}2}S_{1,j_2}y_{1,i_1,\tau_1}H_{\mathbb{Z}1}^{-1}S_{1,j_1}^{-1}I_{\mathbb{Z}1,j_1,j_2,j_3,j_4}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}{\sqrt{6}}$ $-\frac{\hbar\overline{y_{1,i_1,\tau_1}H_{\mathbb{Z}2}S_{1,j_2}y_{1,i_1,\tau_1}H_{\mathbb{Z}1}^{-1}S_{1,j_1}^{-1}I_{\mathbb{Z}1,j_1,j_2,j_3,j_4}^{-1}\lambda_{\mathbb{Z}1}^{-1}}}{\sqrt{6}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	D	F	H
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$W_{\mu\nu}$	constant
T3-D	0	3	0	1	1	2	$\frac{1}{2}$			$-\frac{\hbar\left(W_{\mu_1\mu_2}^A\right)^2}{2\epsilon}$	$\begin{aligned} & +\frac{2\overline{h\overline{y_{1,1},r_1}y_{1,2},r_2}S_{1,j_2}^2y_{1,1},r_2y_{1,2},r_1\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon} \\ & \quad +\text{CN ewScalar}12S_{1,j_1}S_{1,j_1}^\dagger \\ & \quad -\frac{\text{CN ewScalar}12\overline{h\overline{y_{1,1},r_1}S_{1,j_1}y_{1,1},r_1}S_{1,j_1}^\dagger}{\sqrt{3}} \\ & \quad -\frac{2\text{CN ewScalar}12\overline{h\overline{y_{1,1},r_1}S_{1,j_1}y_{1,1},r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon} \\ & \quad -\frac{4h\overline{y_{1,1},r_1}y_{1,2},r_2}S_{1,j_1}S_{1,j_2}y_{1,1},r_2y_{1,2},r_1S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & \quad +\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^1\lambda_{S1}{}^1 \\ & \quad -\frac{2h\overline{y_{1,1},r_1}S_{1,j_3}S_{1,j_4}y_{1,1},r_1S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^1\lambda_{S1}{}^1}{\sqrt{3}\epsilon} \\ & \quad -\frac{h\overline{y_{1,1},r_1}S_{1,j_3}S_{1,j_4}y_{1,1},r_1S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^1\lambda_{S1}{}^1}{\sqrt{3}} \\ & \quad +\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^2\lambda_{S1}{}^2 \\ & \quad -\frac{2h\overline{y_{1,1},r_1}S_{1,j_3}S_{1,j_4}y_{1,1},r_1S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^2\lambda_{S1}{}^2}{\sqrt{3}\epsilon} \\ & \quad -\frac{h\overline{y_{1,1},r_1}S_{1,j_3}S_{1,j_4}y_{1,1},r_1S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S1,j_3,j_1,j_4,j_2}{}^2\lambda_{S1}{}^2}{\sqrt{3}} \end{aligned}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	$W_{\mu\nu}$	constant
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	$B_{\mu\nu}$	F	H	F	$W_{\mu\nu}$	H	e
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$B_{\mu_1\mu_2}^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$B_{\mu_1\mu_2}^2$	$-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$B_{\mu_1\mu_2}^2$	$-\frac{3}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{3}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{3}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)$	$\left(W_{\mu_1\mu_2}^{A_1}\right)^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F H$	$F W_{\mu\nu}$	$H e$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right) B_{\mu_1\mu_2}^2$	$+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}}L_{\tau_1\tau_2}{}^T C P_R L_{\tau_3\tau_4} H_{\tau_1} H_{\tau_2} \lambda_{\tau_3\tau_4} \text{Bar}(\text{SU}(L))_{\tau_1\tau_3} \text{Bar}(\text{SU}(L))_{\tau_2\tau_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} C P_R L_{\tau_3\tau_4}{}^T W_{\tau_1\tau_2}{}^T C P_R L_{\tau_3\tau_4} H_{\tau_1} H_{\tau_2} \lambda_{\tau_3\tau_4} \text{Bar}(\text{SU}(L))_{\tau_1\tau_3} \text{Bar}(\text{SU}(L))_{\tau_2\tau_4} H_{\tau_1}{}^T H_{\tau_2}{}^T}{\sqrt{3} M_F}$ $+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T S_{\tau_1\tau_2}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T S_{\tau_1\tau_2}{}^T S_{\tau_3\tau_4}{}^T \lambda_{\tau_1\tau_2}{}^T \lambda_{\tau_3\tau_4}{}^T}{\sqrt{3}}$	$-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}}{4e}P_R e_{i_1} H_{A_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}}P_R e_{i_1} H_{A_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}}{4e}P_R e_{i_1} H_{A_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{3}{8}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}P_R e_{i_1} H_{A_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}P_L L_{A_1,i_1} y_{1,i_3,r_1} H_{A_1}{}^\dagger}{4e}$ $+\frac{3}{8}\sqrt{3\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}}P_L L_{A_1,i_1} y_{1,i_3,r_1} H_{A_1}{}^\dagger$ $+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}P_L L_{A_1,i_1} y_{2,i_3,r_1} H_{A_1}{}^\dagger}{4e}$ $+\frac{3}{8}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}P_L L_{A_1,i_1} y_{2,i_3,r_1} H_{A_1}{}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right) B_{\mu_1\mu_2}^2$	$+\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}}L_{\tau_1\tau_2}{}^T C P_R L_{\tau_3\tau_4} H_{\tau_1} H_{\tau_2} \lambda_{\tau_3\tau_4} \text{Bar}(\text{SU}(L))_{\tau_1\tau_3} \text{Bar}(\text{SU}(L))_{\tau_2\tau_4}}{M_F}$ $+\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} C P_R L_{\tau_3\tau_4}{}^T W_{\tau_1\tau_2}{}^T C P_R L_{\tau_3\tau_4} H_{\tau_1} H_{\tau_2} \lambda_{\tau_3\tau_4} \text{Bar}(\text{SU}(L))_{\tau_1\tau_3} \text{Bar}(\text{SU}(L))_{\tau_2\tau_4} H_{\tau_1}{}^T H_{\tau_2}{}^T}{M_F}$ $+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $+\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T \text{Bar}(\text{FIM}(SU(3)))_{\tau_1\tau_3} \text{Bar}(\text{FIM}(SU(3)))_{\tau_2\tau_4} H_{\tau_1}{}^T H_{\tau_2}{}^T}{\sqrt{3}}$ $+\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T \text{Bar}(\text{FIM}(SU(3)))_{\tau_1\tau_3} \text{Bar}(\text{FIM}(SU(3)))_{\tau_2\tau_4} H_{\tau_1}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $+\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T \text{Bar}(\text{FIM}(SU(3)))_{\tau_1\tau_3} \text{Bar}(\text{FIM}(SU(3)))_{\tau_2\tau_4} H_{\tau_1}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T S_{\tau_1\tau_2}{}^T S_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $+2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T S_{\tau_1\tau_2}{}^T S_{\tau_3\tau_4}{}^T \lambda_{\tau_1\tau_2}{}^T \lambda_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T \lambda_{\tau_1\tau_2}{}^T \lambda_{\tau_3\tau_4}{}^T}{\sqrt{3}}$ $+\frac{2\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)}{\sqrt{3}}\frac{\overline{W_{\tau_1\tau_2}W_{\tau_3\tau_4}} H_{\tau_1} S_{\tau_1\tau_2}{}^T W_{\tau_3\tau_4}{}^T Y_{\tau_1\tau_2} H_{\tau_3}{}^T S_{\tau_3\tau_4}{}^T \lambda_{\tau_1\tau_2}{}^T \lambda_{\tau_3\tau_4}{}^T}{\sqrt{3}}$	$-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}}{4e}P_R e_{i_1} H_{A_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}}P_R e_{i_1} H_{A_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}}{4e}P_R e_{i_1} H_{A_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}}P_R e_{i_1} H_{A_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}P_L L_{A_1,i_1} y_{1,i_3,r_1} H_{A_1}{}^\dagger}{4e}$ $+\frac{3}{8}\sqrt{3\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}}P_L L_{A_1,i_1} y_{1,i_3,r_1} H_{A_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}P_L L_{A_1,i_1} y_{2,i_3,r_1} H_{A_1}{}^\dagger}{4e}$ $+\frac{3}{8}\sqrt{3\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}}P_L L_{A_1,i_1} y_{2,i_3,r_1} H_{A_1}{}^\dagger$

Field combinations 4 of 6

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	H
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			
T3-B	0	2	0	2	1	1	$\frac{1}{2}$			
T3-C	0	2	0	2	1	3	$\frac{1}{2}$			
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \mathcal{I}_{\tau 2,a_1,a_2,j_1} \lambda_{\tau 2} S_2^\dagger}{\sqrt{3}}$		
								$-h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \mathcal{I}_{\tau 2,a_1,a_2,j_1} \lambda_{\tau 2} S_2^\dagger$		
								$-2h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{e,i_1,\tau_1}} \overline{y_{e,i_2,\tau_2}} H_{a_1} S_2 y_{2,i_2,\tau_2} y_{e,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger$		
								$-h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 2} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$		
								$-h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \lambda_{\tau 2} S_2 y_{2,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2h \log \left(\frac{\mu \text{bar} 2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{\text{H} 2,a_1,a_2} \lambda_{\text{H} 2}^{-1}$		